



Amazon ML Summer School – 2026

Previous Year Interview Experiences

Reading previous participants' experiences is one of the best ways to understand the selection process, exam difficulty, preparation strategy, and internship opportunities.

Recommended Resources

1. Amazon ML Summer School 2023 Experience and Journey to Internship: [1st Interview Experience](#)
 2. Making the Most of ML Summer School 2023: [2st Interview Experience](#)
 3. Amazon Applied Scientist Intern Interview Experience: [3rd Interview Experience](#)
 4. From ML Summer School to Amazon Applied Scientist Intern: [4th Interview Experience](#)
 5. Amazon ML Summer School 2024 Experience: [5th Interview Experience](#)
 6. Amazon ML Summer School Selection Journey: [6th Interview Experience](#)
 7. Amazon ML Summer School 2023 Experience: [7th Interview Experience](#)
 8. Community Discussion and Preparation Insights: [8th Interview Experience](#)
 9. Amazon ML Summer School'23 Experience: [9th Interview Experience](#)
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Previous Year Question Papers

Solving previous year papers should be a mandatory part of your preparation.

1. Amazon ML Summer School 2025: [Amazon Previous Year Question Paper \[ML Summer School 2025 \]](#)
2. Amazon ML Summer School 2024: [Amazon Previous Year Question Paper \[ML Summer School 2024 \]](#)
3. Amazon ML Summer School 2023: [Amazon Previous Year Question Paper \[ML Summer School 2023 \]](#)

Recommendation: Solve all three papers before the assessment. Many concepts are repeated every year.

Amazon ML Summer School Sample Test: [Amazon ML Summer School Sample Test](#)

Recommendation: Attempt the sample paper first to understand the pattern and expected difficulty level.

Amazon DSA Preparation Resources

Although the assessment is ML-focused, DSA preparation is highly beneficial for internship interviews and future Amazon hiring processes.

Frequently Asked DSA Topics

- Arrays
- Strings
- Hashing
- Sliding Window
- Binary Search
- Linked Lists
- Stacks and Queues
- Trees
- Graphs
- Heaps
- Tries
- Greedy Algorithms
- Dynamic Programming

Amazon Frequently Asked DSA Questions

<https://www.geeksforgeeks.org/explore?page=1&company=Amazon&sortBy=submissions>

Assessment Pattern

MCQ Assessment

- Total Questions: 20
- Question Type: Multiple Choice Questions (MCQs)

Subjects Covered

- Machine Learning Fundamentals
 - Probability
 - Statistics
 - Linear Algebra
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High-Priority Topics (Must Prepare)

1. Probability and Statistics

This is consistently one of the most important sections in the assessment.

Probability Fundamentals

- Conditional Probability
- Bayes Theorem
- Independent Events
- Dependent Events
- Random Variables

Statistical Measures

- Expectation (Mean)
- Variance
- Standard Deviation

Probability Distributions

- Bernoulli Distribution
- Binomial Distribution
- Poisson Distribution
- Normal Distribution

Advanced Concepts

- Central Limit Theorem
- Hypothesis Testing
- p-value
- Confidence Intervals
- Covariance
- Correlation

Frequently Asked Questions

- Difference between Covariance and Correlation
 - Effect of multiplying all values by a constant on Variance
 - Bayes Theorem-based Probability Questions
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2. Linear Algebra

Vector Concepts

- Vectors
- Dot Product
- Vector Projection
- Orthogonality

Matrix Concepts

- Matrix Operations
- Matrix Multiplication
- Identity Matrix
- Inverse Matrix
- Determinant
- Rank of Matrix

Advanced Concepts

- Eigenvalues
- Eigenvectors

Frequently Asked Questions

- Conditions for Matrix Invertibility
 - Properties of Eigenvectors
 - Matrix Dimension Calculations
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3. Machine Learning Fundamentals

This is the most important topic for the Amazon ML Summer School assessment.

Learning Paradigms

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Core Concepts

- Classification vs Regression
- Overfitting
- Underfitting
- Bias-Variance Tradeoff
- Train-Test Split
- Cross Validation
- Feature Engineering
- Feature Selection
- Data Leakage

Important Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Support Vector Machines (SVM)
- K-Means Clustering

Frequently Asked Questions

- Why is Logistic Regression used for Classification?
 - Difference between Bagging and Boosting
 - What causes Overfitting?
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4. Evaluation Metrics

Topics to Prepare

- Accuracy
- Precision
- Recall
- F1 Score
- Specificity
- Confusion Matrix
- ROC Curve
- AUC

Frequently Asked Questions

- When Accuracy becomes misleading
 - Precision vs Recall
 - Interpretation of F1 Score
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5. Deep Learning Basics

Neural Network Fundamentals

- Perceptron
- Artificial Neural Networks
- Forward Propagation
- Backpropagation

Activation Functions

- ReLU
- Sigmoid
- Tanh

Optimization Concepts

- Gradient Descent
- Learning Rate

Training Challenges

- Vanishing Gradient Problem
- Exploding Gradient Problem

Regularization Techniques

- Dropout
- Batch Normalization

Frequently Asked Questions

- Why is ReLU preferred?
 - Purpose of Dropout
 - Impact of a High Learning Rate
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6. Optimization Techniques

Topics to Prepare

- Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-Batch Gradient Descent
- Momentum
- Adam Optimizer
- Learning Rate Scheduling

Frequently Asked Questions

- Advantages of Adam Optimizer
 - SGD vs Batch Gradient Descent
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7. Python for Machine Learning

Python Fundamentals

- Lists
- Tuples
- Dictionaries
- Sets

Essential Libraries

- NumPy Basics
- Pandas Basics

Additional Concepts

- List Comprehensions
- Time Complexity of Common Operations

Frequently Asked Questions

- Mutable vs Immutable Objects
 - Output-Based Python Questions
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8. Data Preprocessing

Topics to Prepare

- Missing Value Handling
- Outlier Detection
- Standardization
- Normalization
- Feature Scaling
- One-Hot Encoding
- Label Encoding

Frequently Asked Questions

- Why Scaling is Important for KNN and SVM
 - Difference Between Normalization and Standardization
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Medium-Priority Topics

Natural Language Processing (NLP)

Topics

- Bag of Words
 - TF-IDF
 - Word Embeddings
 - Tokenization
 - Stemming
 - Lemmatization
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Computer Vision

Topics

- CNN Basics
 - Convolution
 - Pooling
 - Filters
 - Stride
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Information Theory

Topics

- Entropy

- Information Gain
 - Gini Index
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Frequently Asked Conceptual Questions

Prepare these concepts thoroughly:

1. Why does Random Forest reduce Overfitting?
 2. Why is Naive Bayes called "Naive"?
 3. When should Precision be preferred over Recall?
 4. What is the Curse of Dimensionality?
 5. Why does KNN become slow for large datasets?
 6. What is Data Leakage?
 7. Difference between L1 and L2 Regularization?
 8. Why is Logistic Regression considered a Linear Model?
 9. What is the role of Activation Functions?
 10. Why do Decision Trees overfit?
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